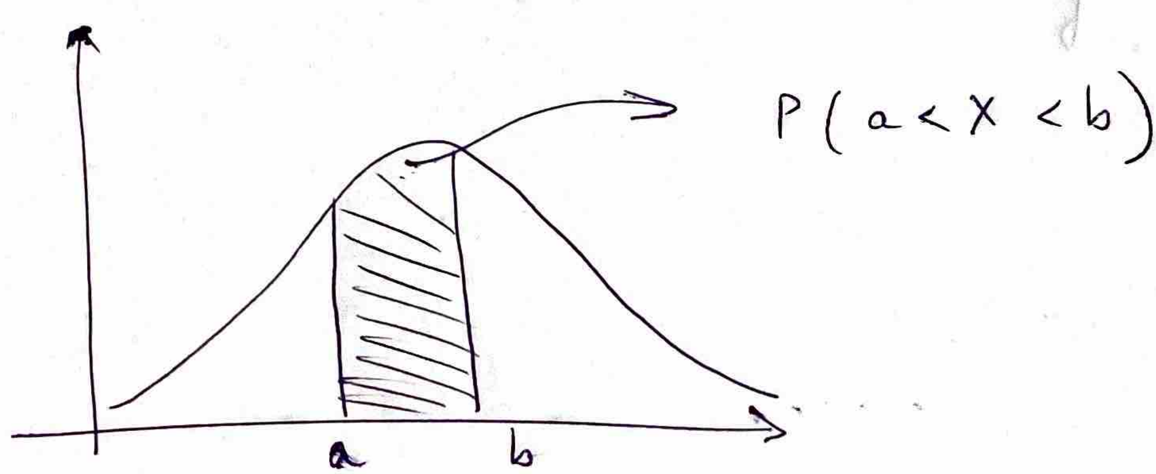
Continuous Probability Distribution

- A continuous random variable has a probability of 0 to assume exactly any of its values. Therefore, it can't be given in tabular form. We don't define probability of 1 particular value being taken by the random variable, assigning it as 0. We talk about intervals here.

- $P(a < X \leq b) = P(a < X < b) + P(X = b)$
$$= P(a < X < b)$$
This explains, it doesn't matter whether we include an endpoint of the interval or not. This is not true in the case of discrete X.

- A probability density function is constructed so that the area under its curve bounded by the x-axis is equal to 1.

$$\boxed{P(a < X < b) = \int_a^b f(x)\, dx}$$



- The function $f(x)$ is a probability density funcⁿ for the continuous random variable X, defined over the set of real nos., if,

1. $f(x) \geq 0, \forall x \in R$
2. $\int_{-\infty}^{\infty} f(x)\, dx = 1$
3. $P(a < X < b) = \int_a^b f(x)\, dx$

- The cumulative distribuⁿ funcⁿ $F(x)$ of a continuous random variable X with density funcⁿ $f(x)$ is -

$$\boxed{F(x) = P(X \leq x) = \int_{-\infty}^{x} f(t)\, dt}, \quad -\infty < x < \infty$$

$$\boxed{P(a < X < b) = F(b) - F(a)}$$

$$f(x) = \frac{dF(x)}{dx} \quad \text{or} \quad F(x) = \int_{-\infty}^{\infty} f(x)\, dx$$

if the derivative exists